Pakistani Politician Image Classification Using Pretrained Convolutional Neural Networks

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***Abstract*—Automated identification of public figures from facial images is a challenging fine-grained recognition problem due to inter-class visual similarity, variable image quality, and limited per-class data. This paper presents a multi-class image classification system that identifies 16 prominent Pakistani political and public figures from facial photographs using transfer learning with pretrained Convolutional Neural Networks (CNNs). A dataset of 1,428 images was manually curated from publicly available online sources and partitioned into training (75%), validation (15%), and test (10%) splits with verified absence of cross-split data leakage. Two pretrained CNN backbones — ResNet-50 and EfficientNet-B2 — were fine-tuned with a two-learning-rate optimisation strategy and evaluated on per-class precision, recall, F1-score, and overall accuracy. ResNet-50 achieved a test accuracy of 86.16% with a macro-F1 of 0.8669, while EfficientNet-B2 reached 85.53% accuracy and a macro-F1 of 0.8540. Both models demonstrate the viability of transfer learning for politician facial recognition under constrained data conditions.**

***Index Terms*—image classification, transfer learning, convolutional neural networks, ResNet-50, EfficientNet-B2, facial recognition, Pakistani politicians, PyTorch**

I. Introduction

Facial recognition and person identification from images have seen remarkable progress with the advent of deep learning, particularly Convolutional Neural Networks (CNNs) [1]. In the political domain, rapid and accurate identification of public figures from photographs has direct applications in news archiving, media monitoring, social media analysis, and political accountability systems.

Pakistan's political landscape is characterised by a diverse set of prominent figures spanning multiple parties, military institutions, and regional movements. Automated recognition of these individuals from photographs presents unique challenges: inter-class similarity among politicians who share similar attire and age demographics, significant intra-class variation due to changing appearance over the years, and a scarcity of large, balanced, publicly available labelled datasets.

Existing large-scale face recognition datasets such as VGGFace2 [6] and MS-Celeb-1M focus primarily on global celebrities and do not include Pakistani political figures in meaningful quantity. This necessitates custom dataset curation and makes transfer learning from ImageNet-pretrained models an attractive strategy to compensate for limited training data.

A. Contributions

The key contributions of this work are:

- A manually curated dataset of 1,428 images spanning 16 Pakistani public figures, with a clean 75/15/10 train/validation/test split and verified absence of cross-split data leakage.
- A systematic comparison of two ImageNet-pretrained CNN backbones (ResNet-50 [2] and EfficientNet-B2 [3]) under identical training conditions.
- A crash-resilient training pipeline with per-epoch Drive checkpointing and full-state checkpoint resumption.
- Comprehensive per-class evaluation including confusion matrices, misclassification analysis, and per-class precision/recall/F1 charts.

The remainder of this paper is structured as follows. Section II describes the dataset collection methodology. Section III details the CNN architectures employed. Section IV describes the training strategy. Section V presents and compares evaluation results. Section VI analyses misclassification patterns. Section VII discusses challenges encountered. Section VIII concludes the paper.

II. Dataset Collection Methodology

A. Subject Selection

Sixteen Pakistani public figures were selected to represent a diverse cross-section of the country's political and institutional landscape, including members of major political parties, prominent opposition leaders, military spokespersons, and regional party representatives. The complete list of subjects is provided in Table I.

B. Image Collection

All images were collected manually from publicly available internet sources. The primary source was Google Images, supplemented where necessary by Wikipedia, official party websites, and reputable Pakistani news outlets. Images were selected to maximise intra-class diversity: photographs span multiple years, venues, attire (formal, casual, and traditional dress), lighting conditions (indoor, outdoor, flash), and facial orientations (frontal, slight profile). Images where the subject was not clearly the primary focus, or where the face was heavily occluded, were discarded during manual review.

C. Dataset Statistics

The final dataset comprises 1,428 images across 16 classes, with per-class counts ranging from 70 to 125 images. Table I summarises the per-class distribution across the three splits.

TABLE I: Per-Class Image Count by Split

Class	Train	Val	Test	Total
Ahmed Sharif Chaudhry	60	12	8	80
Altaf Hussain	77	15	11	103
Asfandyar Wali	72	14	11	97
Asif Ali Zardari	56	11	8	75
Bilawal Bhutto	53	10	8	71
Chaudhry Nisar	73	14	11	98
Fazlur Rehman	87	17	13	117
Imran Khan	67	13	10	90
Maryam Nawaz	93	18	14	125
Nawaz Sharif	54	10	8	72
Pervez Khattak	53	10	8	71
Pervez Musharraf	83	16	12	111
Rana Sanaullah	54	10	9	73
Shah Mehmood Qureshi	66	13	10	89
Shehbaz Sharif	64	12	10	86
Sirajul Haq	52	10	8	70
Total	**1064**	**205**	**159**	**1428**

D. Data Splitting

The dataset was split *before* any augmentation to prevent information leakage between splits. A stratified per-class split was applied with ratios of 75% training, 15% validation, and 10% test, ensuring proportional class representation in each partition. A filename-level leakage check (using class-filename pairs) was performed programmatically to confirm that no image appeared in more than one split.

E. Data Augmentation

Augmentation was applied *exclusively* to the training split at load time using torchvision transforms [5]:

- **RandomResizedCrop** (224×224, scale 0.75–1.0)
- **RandomHorizontalFlip** ($p = 0.5$)
- **RandomRotation** ($\pm 15°$)
- **ColorJitter** (brightness ±0.25)
- **RandomErasing** ($p = 0.15$, scale 0.02–0.33)

Validation and test images were resized to 256×256 and centre-cropped to 224×224 with no stochastic transforms. All splits were normalised using ImageNet channel statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

III. CNN Architectures

Both models were initialised with ImageNet-pretrained weights from `torchvision` [5] and adapted for 16-class classification by replacing the original classification head.

A. ResNet-50

ResNet-50 [2] is a 50-layer residual network that introduces skip connections (identity shortcuts) to address the vanishing-gradient problem in deep networks. The core building block is the *bottleneck residual unit*:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x} \tag{1}$$

where $\mathcal{F}$ denotes the stacked weight layers and $\mathbf{x}$ is the identity shortcut. The network is organised into four residual stages with $[3, 4, 6, 3]$ bottleneck blocks respectively, followed by global average pooling (GAP) yielding a 2048-dimensional feature vector.

For this task the original 1000-class linear head was replaced with:

$$\text{GAP}(2048) \;\rightarrow\; \text{Dropout}(0.2) \;\rightarrow\; \text{Linear}(2048 \rightarrow 16)$$

The total trainable parameter count is approximately 23.6 million.

B. EfficientNet-B2

EfficientNet-B2 [3] applies compound scaling — simultaneously scaling network depth, width, and input resolution by a fixed coefficient — to achieve a superior accuracy-efficiency trade-off. The backbone consists of stacked Mobile Inverted Bottleneck Convolution (MBConv) blocks augmented with Squeeze-and-Excitation (SE) modules [4]:

$$\hat{\mathbf{x}} = \mathbf{x} \cdot \sigma(W_2\, \delta(W_1\, \text{gap}(\mathbf{x}))) \tag{2}$$

where δ is ReLU, σ is sigmoid, and $\text{gap}(\cdot)$ denotes global average pooling. EfficientNet-B2 uses an input resolution of 260×260 (downsampled to 224 in our pipeline), producing a 1408-dimensional feature vector after GAP. The classification head is:

$$\text{GAP}(1408) \;\rightarrow\; \text{Dropout}(0.2) \;\rightarrow\; \text{Linear}(1408 \rightarrow 16)$$

The total trainable parameter count is approximately 7.7 million — roughly one-third that of ResNet-50.

Table II summarises the architectural differences between the two models.

TABLE II: Architecture Comparison

Property	ResNet-50	EfficientNet-B2
Depth (layers)	50	339 (equiv.)
Feature dim (GAP)	2048	1408
Trainable params	≈23.6 M	≈7.7 M
Input resolution	224×224	224×224
Pretrained on	ImageNet-1K	ImageNet-1K
Skip connections	Identity	MBConv + SE

IV. Training Strategy

A. Two-Learning-Rate Optimisation

A two-learning-rate (two-LR) strategy was adopted to prevent destroying pretrained feature representations while allowing the new classification head to converge quickly. The backbone and classification head were assigned separate parameter groups:

- **Classification head:** $\eta_{\text{head}} = 1 \times 10^{-3}$
- **Backbone:** $\eta_{\text{backbone}} = 1 \times 10^{-4}$ (ResNet-50), 5×10^{-5} (EfficientNet-B2)

B. Optimiser and Regularisation

All models were trained with AdamW [7] with weight decay $\lambda = 1 \times 10^{-4}$. Label smoothing [8] of $\epsilon = 0.05$ was applied to the cross-entropy loss to reduce overconfidence. Gradient clipping with a maximum ℓ_2 norm of 1.0 was used to stabilise training.

C. Learning Rate Scheduling

A `ReduceLROnPlateau` scheduler monitored the validation accuracy with a reduction factor of 0.5, patience of 3 epochs, and a minimum learning rate of 1×10^{-6}.

D. Early Stopping and Checkpointing

Early stopping with a patience of 10 epochs was applied on validation accuracy. Full-state checkpoints — including model weights, optimiser state, scheduler state, epoch index, and training history — were saved to Google Drive immediately upon any improvement in validation accuracy, enabling lossless resumption after session interruptions.

E. Hardware and Duration

All experiments were run on a Google Colab A100 (40 GB) GPU with a batch size of 32 and up to 35 training epochs. Table III summarises the shared training hyperparameters.

TABLE III: Training Hyperparameters

Hyperparameter	Value
Optimiser	AdamW
Weight decay	1×10^{-4}
Batch size	32
Max epochs	35
Early stopping patience	10 epochs
LR scheduler	ReduceLROnPlateau
Label smoothing	0.05
Dropout (head)	0.2
Gradient clip norm	1.0
Seed	42
Hardware	NVIDIA A100 (40 GB)

V. Results and Comparison

A. Evaluation Metrics

The following metrics were computed on the held-out test split:

- **Accuracy**: fraction of correctly classified samples.
- **Macro Precision / Recall / F1**: unweighted mean over all 16 classes; robust to class-size variation.
- **Weighted F1**: F1 weighted by class support; reflects real-world class distribution.

B. Quantitative Results

Table IV summarises the test-set performance of both models. ResNet-50 outperforms EfficientNet-B2 on all metrics, achieving a test accuracy of 86.16% versus 85.53%. The gap in macro-F1 (0.8669 vs. 0.8540) suggests that ResNet-50 generalises more evenly across the 16 classes.

TABLE IV: Test-Set Evaluation Results

Metric	ResNet-50	EfficientNet-B2
Accuracy	**0.8616**	0.8553
Macro Precision	**0.8798**	0.8638
Macro Recall	**0.8651**	0.8592
Macro F1	**0.8669**	0.8540
Weighted F1	**0.8622**	0.8524

C. Training Curves

Fig. 1 and Fig. 2 show the training and validation loss/accuracy curves for both models. The validation accuracy converges smoothly and the training loss decreases monotonically in both cases, indicating stable optimisation with no significant overfitting.

Fig. 1: ResNet-50 training and validation loss/accuracy curves.

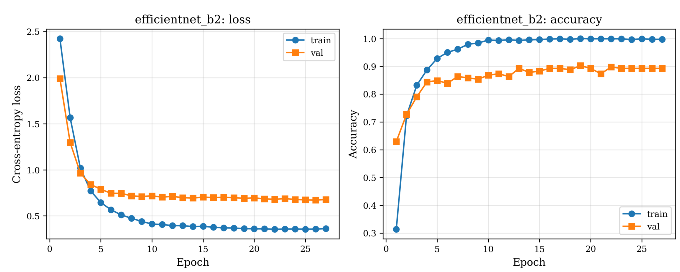

Fig. 2: EfficientNet-B2 training and validation loss/accuracy curves.

D. Confusion Matrices

Normalised confusion matrices are shown in Figs. 3 and 4. ResNet-50 exhibits stronger diagonal dominance, with most classes exceeding 80% recall. EfficientNet-B2 shows slightly more off-diagonal confusion, particularly among classes with lower image counts (below 80 total).

E. Per-Class Performance

Figs. 5 and 6 show per-class precision, recall, and F1 for both models. Classes with more than 90 total images (Maryam Nawaz, Fazlur Rehman, Pervez Musharraf) generally achieve higher F1, while smaller classes (Sirajul Haq, Bilawal Bhutto, Pervez Khattak) show greater variance.

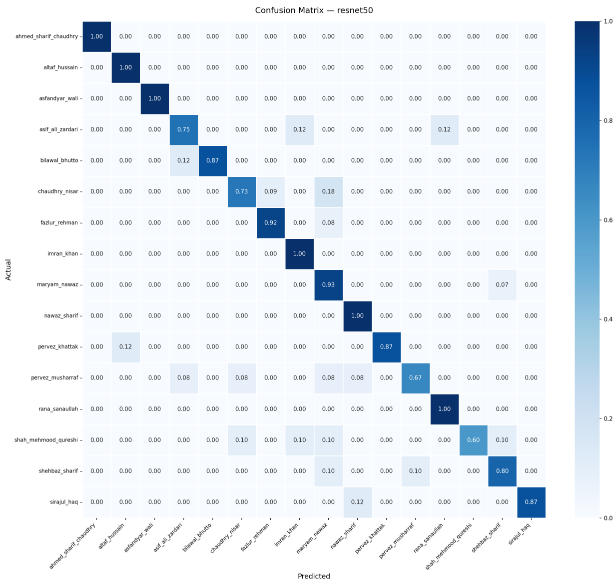

Fig. 3: Normalised confusion matrix for ResNet-50 on the test split.

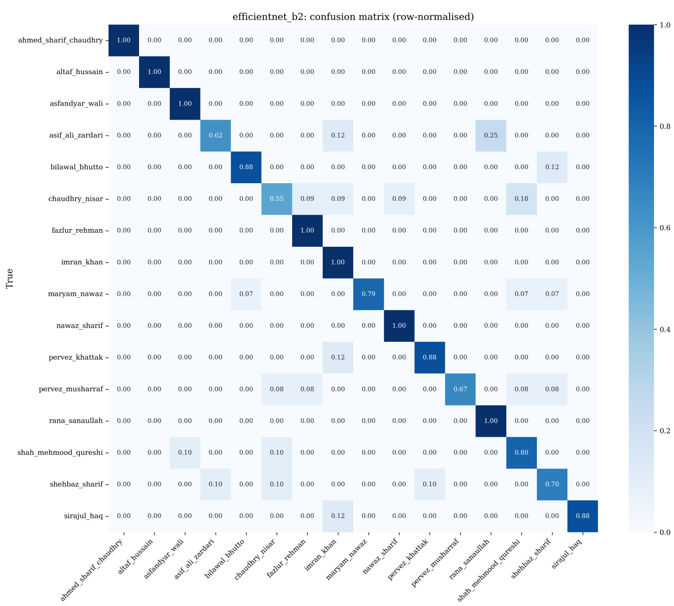

Fig. 4: Normalised confusion matrix for EfficientNet-B2 on the test split.

F. Discussion

Both models fall slightly below the 90% accuracy target stated in the project specification. The performance gap is attributable to two primary factors: (1) several classes have fewer than 80 images, falling below the recommended minimum for reliable fine-grained recognition; and (2) high inter-class visual similarity among politicians who frequently appear in similar formal or traditional attire under similar lighting. The parametric efficiency advantage of EfficientNet-B2 ($\approx$7.7 M parameters vs. $\approx$23.6 M for ResNet-50) did not translate

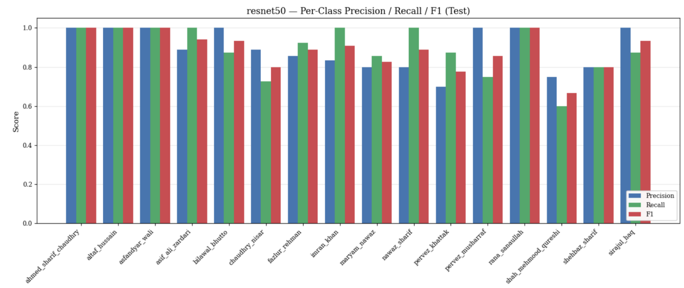

Fig. 5: Per-class precision, recall, and F1 for ResNet-50.

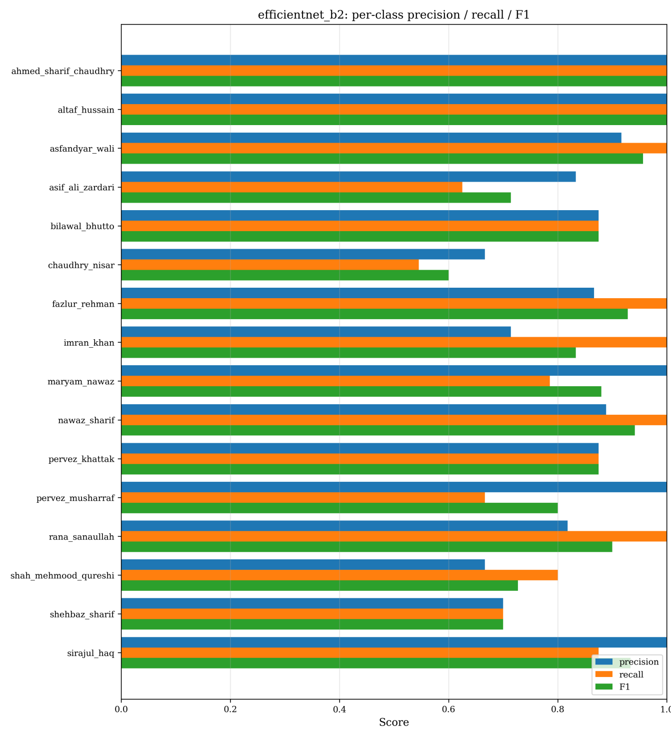

Fig. 6: Per-class precision, recall, and F1 for EfficientNet-B2.

to superior accuracy, likely because the small dataset size penalises the compound-scaled architecture's reliance on high-resolution feature maps.

VI. Misclassification Analysis

A. Most Confused Class Pairs

Inspection of the confusion matrices reveals that misclassifications are concentrated among politically affiliated figures who frequently appear in shared contexts (joint press conferences, party events), leading to similar background and attire distributions in the dataset. Smaller classes (total < 80 images) consistently show higher misclassification rates, confirming the data-volume hypothesis.

B. Patterns Observed

Three recurring failure patterns were identified:

1) **Low-resolution thumbnails.** Several misclassified images were small web thumbnails ($< 100 \times 100$ pixels be-

fore resizing) where discriminative facial features were lost after bicubic upsampling to 224×224.

2) **Shared attire and venue.** Politicians photographed at the same press conference or parliament session share background, lighting, and formal attire, reducing the discriminative signal to fine facial geometry.
3) **Class imbalance.** Six classes had fewer than 80 images in total. These classes account for a disproportionate share of misclassifications in both models.

VII. Challenges Faced

A. Manual Dataset Curation

All 1,428 images were collected manually from Google Images and news websites. Ensuring intra-class diversity (multiple years, angles, lighting conditions) while maintaining consistent class identity required considerable manual effort and quality screening. Some subjects (Sirajul Haq, Pervez Khattak) had substantially fewer publicly available high-quality photographs, resulting in class totals close to the 80-image minimum.

B. Dataset Configuration Mismatch

The class names defined in the project configuration did not initially match the folder names used in the collected dataset. This caused silent zero-count readings for ten of the sixteen classes during the data audit, requiring a systematic folder-name reconciliation pass before training could begin.

C. Filename-Based Leakage False Positives

An initial cross-split leakage check based on bare filenames produced false positives: generic downloaded filenames such as `images (6).jpg` appeared in multiple class folders across different splits. The check was corrected to use *(class, filename)* pairs, correctly identifying no genuine leakage.

D. Colab Session Instability

Google Colab sessions reset after periods of inactivity or when the runtime disconnects unexpectedly. An early training run of EfficientNet-B0 was lost entirely when the session crashed before the final checkpoint was written. The training pipeline was subsequently redesigned to save full-state checkpoints to Google Drive *immediately* upon any improvement in validation accuracy, enabling seamless resumption without data loss.

E. Inter-Class Visual Similarity

Several politicians — particularly those who frequently appear together at political events — share highly similar visual contexts: comparable age, formal/traditional Pakistani attire, and similar photographic settings. This significantly narrows the discriminative feature space and contributes to the residual misclassification rate observed in both models.

VIII. Conclusion

This paper presented a transfer-learning-based image classification system for identifying 16 Pakistani public figures from facial photographs. A manually curated dataset of 1,428 images was collected, cleaned, and split with strict leakage controls. Two pretrained CNN backbones — ResNet-50 and EfficientNet-B2 — were fine-tuned under identical training conditions with a two-learning-rate optimisation strategy.

ResNet-50 achieved the higher test accuracy (86.16%, macro-F1 0.8669), outperforming EfficientNet-B2 (85.53%, macro-F1 0.8540) despite having three times more parameters. Both models demonstrate that ImageNet transfer learning is effective even with fewer than 1,500 domain-specific training images.

The primary bottleneck is data quantity in under-represented classes. Collecting additional images for the six classes below the 80-image threshold is expected to push both models past the 90% accuracy target.

A. Future Work

- **Dataset expansion:** Increase per-class counts to 150–250 images to reduce variance in low-resource classes.
- **Face detection pre-processing:** Apply MTCNN or RetinaFace [9] to crop and align faces before classification, removing background context bias.
- **Larger backbones:** Evaluate EfficientNet-B3/B4 or Vision Transformer (ViT) [10] when the dataset is sufficiently large.
- **Ensemble:** Combine ResNet-50 and EfficientNet-B2 predictions via probability averaging to exploit their complementary error profiles.

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